



**BIG
G EXPRESS**

Big G Express Project

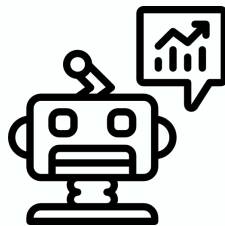
“Highway to ML”

Grant Alan, Abigail Ezell, Cat Gomez, Sarah Mulloy,
and Sivaraja Vaithiyalingam

Table of Contents

- Project Intro
- Data pre-processing
- Feature engineering
 - K-fold validation
- Models and Results
- Imbalanced Class Optimization
 - Feature importance





Project Goals

Truck emission has sensor to detect system faults



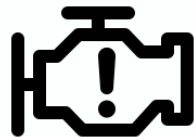
Derate occurs when engine output is limited, resulting in truck service and tow



Predict **likelihood of derate** after a system fault code fast enough ($> 2\text{hrs}$) for a truck to reach a service stop to avoid tow



Dataset Information



Service Fault Codes

- Date and time
- vehicle ID
- Fault ID



Vehicle Diagnostics of Fault Codes

- Equipment failure
- Truck Metrics (pressure, temperature, speed)
- Additional diagnostic settings





Data Pre-Processing



Data prep

- Merged faults data with diagnostic data
- Dropped irrelevant columns and misleading rows



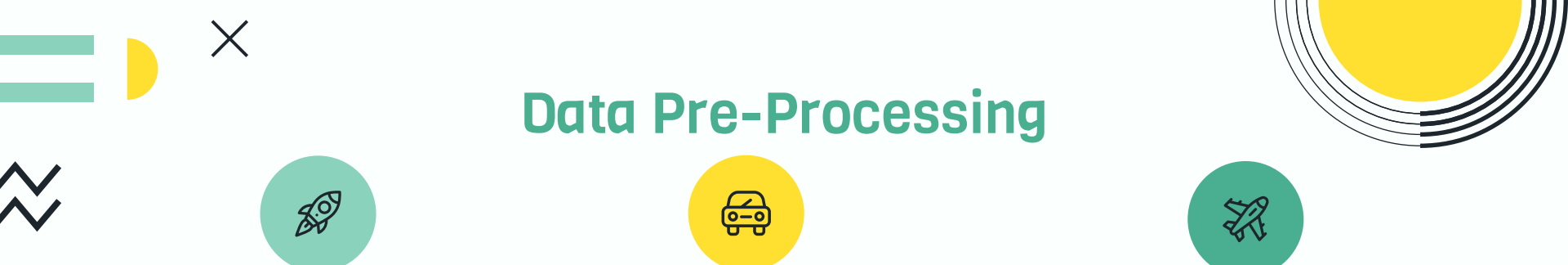
Define Target

- “Will this truck derate in the next __ hours?”
- Three time windows: 6 hrs, 12 hrs, and 24 hrs
- Observations within 2 hours of a derate were excluded from testing



Scaling/Encoding

- Used MaxAbsScaler due to sparse data
 - Used OneHotEncoder for categorical features



Data Pre-Processing



Missing Data

- Forward-filled running counts
- Used IterativeImputer for missing numeric values
- Used SimpleImputer for missing categorical values



Imbalanced Classes

- Upweighted the minority class
- Oversampled the minority class (SMOTE)



Train/test split

- Sorted data chronologically
- Split into train, validation, and test sets



Feature Engineering

From Static Telemetry to Degradation Dynamics



Baseline Operational Predictors



Operational Overlap Between Derate and Non-Derate States



EDA (Exploratory Data Analysis) of Instability and Recovery Behavior



Engineered Predictive Features



Degradation-Dynamics Framework

- Baseline operational predictors showed substantial overlap between derate and non-derate states
- Static telemetry values alone were insufficient for stable early-warning prediction
- Time-based validation revealed poor operational separability under real-world conditions
- EDA identified progressive instability and shrinking recovery behavior approaching future derates
- Feature engineering focused on temporal degradation dynamics rather than isolated abnormal sensor values



Evolution of Predictive-Feature Frameworks

Framework

◆ Baseline Predictors



◆ Rolling Features



Instability Features

DG Features

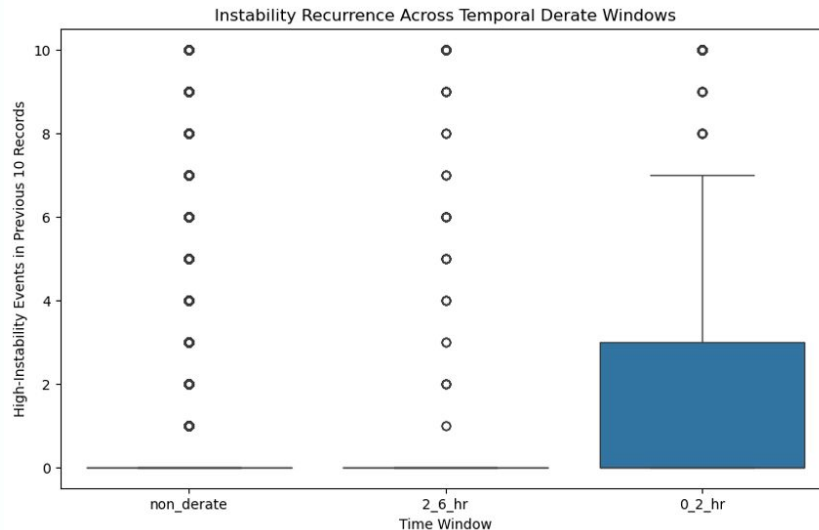
Operational Meaning

Static operational state

Short-term trends

Operational fluctuation

Progressive
degradation dynamics



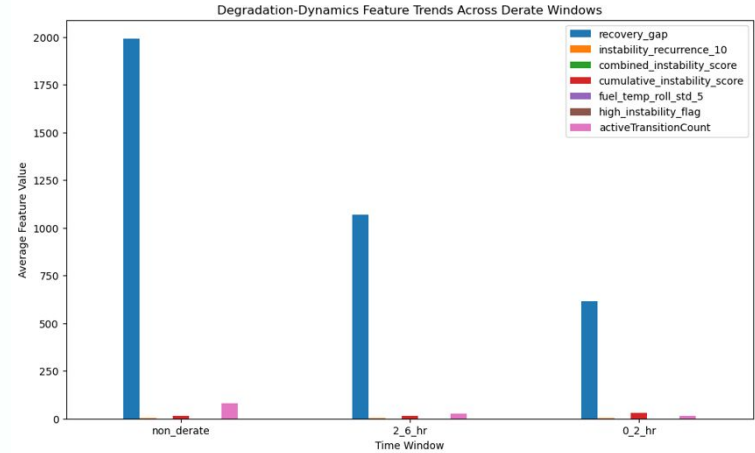
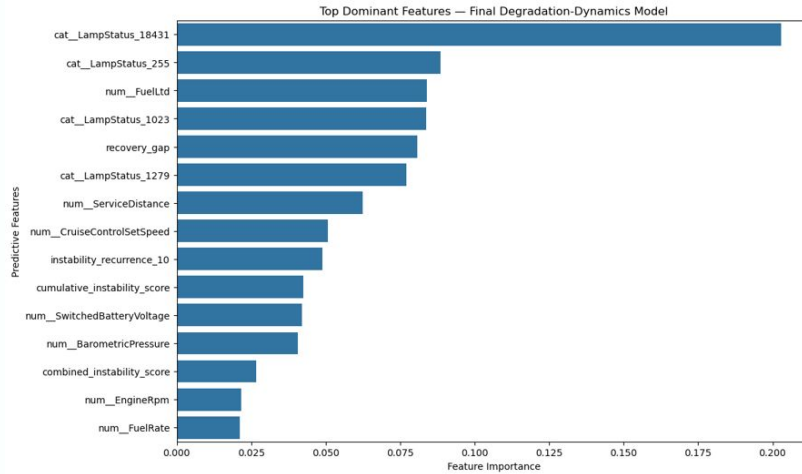
Final Degradation-Dynamics Predictors

- Combined instability score
- Instability recurrence
- Recovery gap behavior
- Instability density
- Recovery-trend dynamics

- Increasing instability recurrence was observed approaching future full derate events
- Trucks with persistent instability frequently demonstrated shrinking operational recovery behavior
- These patterns suggested progressive operational resilience collapse rather than isolated abnormal telemetry states



Evolution of Predictive-Feature Frameworks



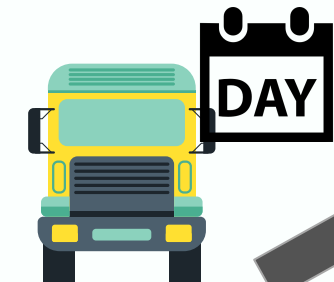
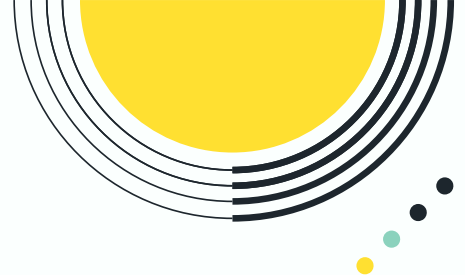
- Recovery-gap behavior emerged as one of the dominant predictors within the final degradation-dynamics framework
- Persistent instability and shrinking recovery periods were strongly associated with future full derate behavior
- Degradation-dynamics features improved temporal degradation separability compared to earlier feature-engineering frameworks
- The strongest degradation signals emerged closer to future full derate events

K-fold Validation for model evaluation





Cost Calculations



False Negative



Predicted derate



Actual derate



\$0
(\$4000)

True Positive



Predicted derate



Actual derate



\$4000

False Positive



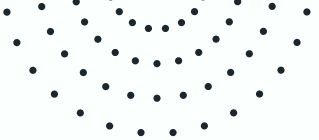
Predicted derate



Actual derate



\$500



Gradient-Boosted Decision Tree Model



Model details:

- XGBoost model
- Predicting derates 6 hours out
- Upweighted minority class using `scale_pos_weight`

Most impactful features (by average shap score):

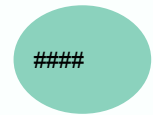
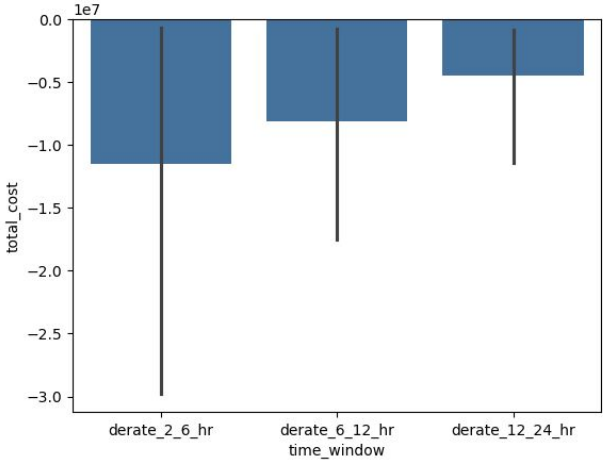
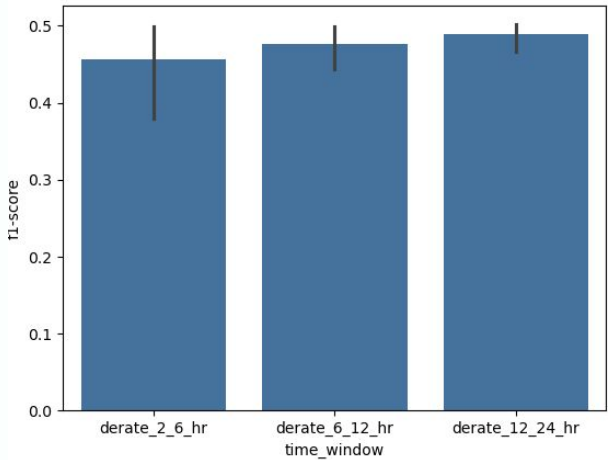
- Engine time (lifetime to date)
 - FMI 3.0
- Distance (lifetime to date)
- Fuel (lifetime to date)
- Minutes since last fault

Real-world performance:

\$2,000 cost

Would have caught 0 derates
6 false positives

Neural Network Classifier



Feature Name



Feature Name



Feature Name



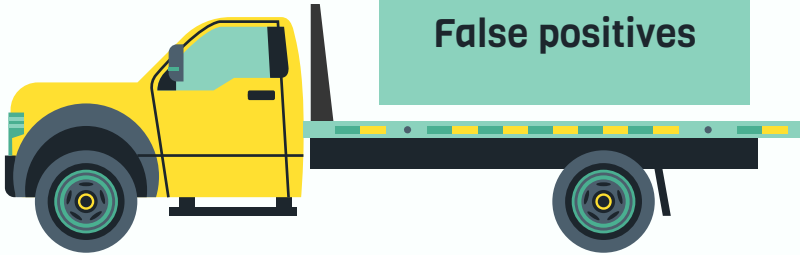
Model Performance

\$416,000

Real-world
performance
\$416,000

Number derates
detected

False positives



Random Forest Classifier



Imbalanced Class Optimization



Massively Imbalance Dataset

Out of 188,771 samples, only 174 (**0.09%**) represent critical derate events, causing our model to be very accurate but relatively useless.

Class Weighting

By shifting the class weight in our model to “balanced”, we force our model to penalize falsely predicted minorities more harshly. Force the model to pay more attention to derates.

Threshold Tuning

Used predicted probabilities instead of a default 0.5 classification cutoff to figure out which threshold amongst our dataset maximized cost savings. We evaluated the model using the optimized threshold but my model was still not useful.

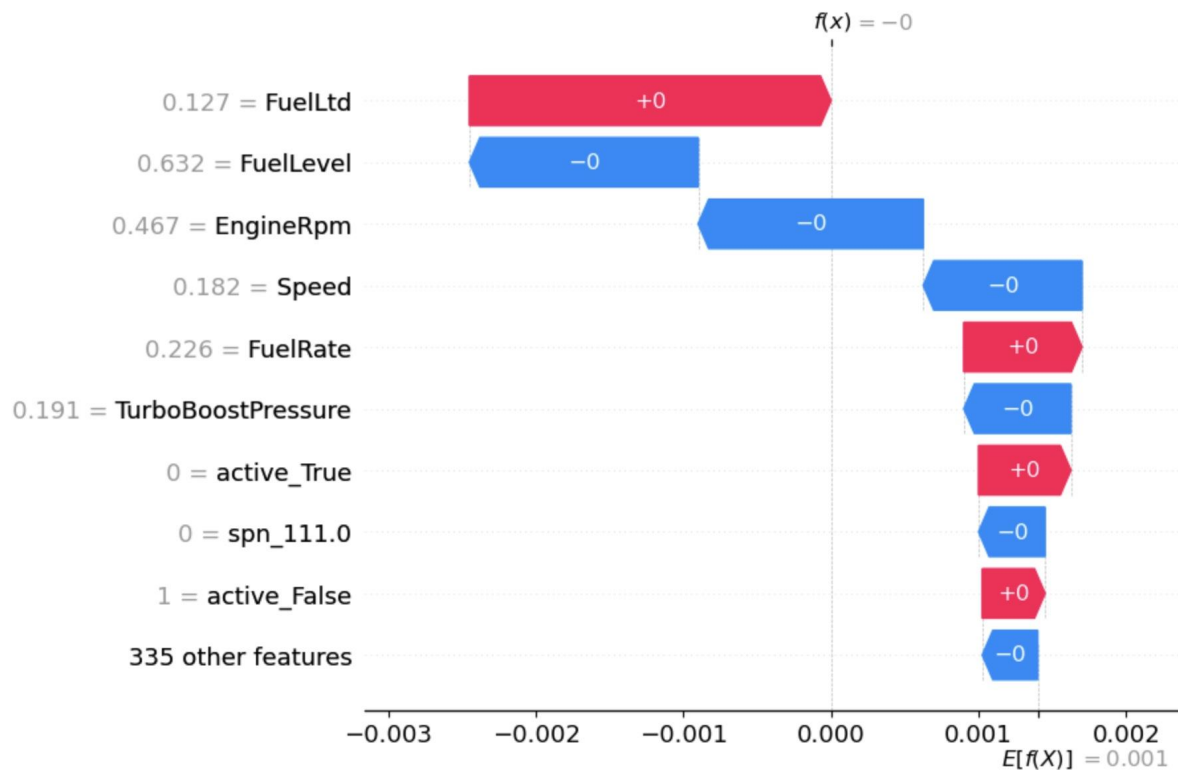
[75] :	threshold	cost_savings
0	0.10	697500
1	0.11	697000
2	0.12	696500
3	0.13	696500
4	0.14	696500
54	0.64	696000
61	0.71	696000
60	0.70	696000
59	0.69	696000
58	0.68	696000

SHAP + Interpretability

The SHAP waterfall plot explains why the model predicted a high or low probability of derate for a specific observation.

FuelLtd, FuelLevel, EngineRpm and Speed have the largest influence on this specific prediction.

Red features are pushing our prediction toward derate and the blue features are doing the opposite. The Blue features decrease derate probability.



It's not the destination,
it's the journey.

Ralph Waldo Emerson

